Krunal Rathod

Driven to push boundaries of intelligent autonomy by integrating AI with engineering for next-generation autonomous systems

Education

2024 ↑ 2022

Master of Science in Artificial Intelligence

Università della Svizzera italiana, Lugano, Switzerland

Thesis and Achievements:

- **Master Thesis**: Computer Vision and Any-Angle Path Planning Algorithm for a Training App on Sprint Orienteering Maps, **ETH Zürich**, 2024.
  - Combined Computer Vision and AI-based path planning to develop a training app for Sprint Orienteering Maps, using CNN for feature map extraction and any angle path planning algorithms, tested on different algorithms, including A-star, Theta-star, RRT, DQN, ACS, alongside TSP-inspired metaheuristics for optimality, scalability, and adaptability
  - Including testing on simulated and real maps, and iOS app development for performance evaluation based on path accuracy and user feedback.
- Awarded **60 One-off Masters Scholarship**.

2025 ↑ 2020

Masters of Technology in Mechanical Engg.

Marwadi University, Rajkot, India

Thesis and Achievements:

- Specialization in Computer Aided Design / Computer Aided Manufacturing (CAD/CAM)
- **Master Thesis**: Computer Vision based object recognition pick and place robotic arm using TensorFlow, 2022.
  - Developed 6DoF Kinova Arm using ROS (Noetic) in Unity and Gazebo for control and validation
  - Applied computer vision techniques, utilizing YOLOv4 and SSD MobilenetV2 for agricultural and waste sorting application, ensured practical utility for addressing challenges in integrating robotics and computer vision.
- Maintained **1st Rank in CAD/CAM Specialisation**

2020 ↑ 2017

Bachelor of Engineering in Automobile

Gujarat Technological University, Ahmedabad, India

Thesis and Achievements:

- **Bachelor Thesis**: All Terrain Hybrid Lawnmower, 2020.
  - As a part of team, design and developed a lawnmower, integrating mechanical, electrical and control system to enable operation on diverse terrains
  - Helped in Simulations and prototyping the model in Siemens NX, to optimize hybrid power system and terrain handling capabilities
- Awarded **1st Place in Education** by Merchant Engineering College, affiliated to GTU.

2017 ↑ 2014

Diploma Engineering in Automobile

Gujarat Technological University, Ahmedabad, India

Technical Seminar:

- Presentation on Satellite Aided Transmission in Rolls-Royce Phantom, *at Sir BPTI, Bhavnagar*, 2017.

Contact

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Skills

- **System**: Linux, MacOS, Windows
- **Programming**: C/C++, Python, MATLAB, Java, Swift
- **Tools & Libraries**: TensorFlow, PyTorch, OpenCV, ROS/ROS2, Scikit Learn, PCL, LaTeX
- **DevOps & Version Control**: Git, Docker, Visual Studio, Android Studio, XCode
- **Technical**: Gazebo, CoppeliaSim, MuJoCo, Unity, AutoDesk Fusion 360, Siemens NX, ANSYS

Languages

English	Advance
Gujarati	Fluent
Hindi	Fluent
Urdu	Advance
German	Basic

Work experience

present ↑ April 2023

Volunteer (IT Team), *(2 yr)*

voCHabular, Zürich, Switzerland

- Developed website to support learning Swiss and Standard German, creating platform using web technologies to enhance user engagement in language learning
- Developed a dictionary app for multiple languages, focusing on Swiss and Standard German, integrating database system and NLP for deliver accurate translation
- Enhancing Transformer-based chatbot to enable conversational learning of Swiss and Standard German, employing deep learning techniques to facilitate natural, context-aware dialogues

March 2024 ↑ March 2023

EPFL AI Team, *(1 yr)*

EPFL AI Team, EPFL, Laussane, Switzerland

- Served as a key liaison between EPFL AI Team and industry representatives
- Contacted and Initiated EPFL relationship with industry leaders in AI to establish partnerships, securing opportunities to enhance academia and industry connection
- Organized and facilitated meetings with industry representatives to explore mutual interests, such as research, placement, events, etc.

Dec 2021 ↑ Dec 2020

Data Analyst Intern, *(1 yr)*

J2ML InfoTech Pvt. Ltd., Ahmedabad, India

- Developed and optimized machine learning algorithms for predictive analytics, achieving 90% accuracy in forecasting healthcare outcomes and user behavior on job portals
- Implemented NLP pipelines to classify and analyze user reviews, enhancing data-driven decision making
- Conducted exploratory data analysis using Python, Pandas, and NumPy to identify key features, improving model performance and business insights
- Collaborated with cross functional teams and integrated models into real-time applications, ensuring scalability and usability

May 2019 ↑ July 2018

Coursework Support, *(11 mos)*

Merchant Engineering College, Mehsana, India

- Assisted in delivering advanced coursework during 5th and 6th semesters, including, Automobile Transmission, Automobile Systems, Automotive Computer Controlled Systems, Vehicle Dynamics, by supporting lab experiments, simulation and documentation
- Support creation of lab manuals and assignments, fostering technical skills in system integration and analysis for interdisciplinary research

Sept 2017 ↑ June 2017

Assi. Warehouse Manager, *(4 mos)*

AMW Ltd, Bhuj, India.

- Managed receiving, inspection, and inventory control for a high-volume production assembly line, ensuring accurate and timely material availability
- Coordinated material flows across procurement, warehouse, and production teams, optimizing logistics to maintain seamless operation
- Utilized inventory management software to track stock level, generate reports, and forecast a material need, enhancing operation efficiency

University Projects

- **Snooker Table 3D Reconstruction (Computer Vision & Pattern Recognition)**: Preprocessed video frames to match with a specified reference frame of the table using pixel similarity. Computed Camera Matrix using the DLT to decompose into Calibration Matrix,Orientation and Center. Detected ball position in individual frames and projected them to a top–down view using homography

- **Trash Detection & Collection Thymio Robot (Robotics)**: Developed an autonomous trash detection thymio robots with ROS2. Utilized PyTorch and OpenCV to train and deploy a YOLOv5 model for garbage detection, interfacing it with robot. Employed in CoppeliaSim to simulate and test the behavior in virtual environment.

- **Reproduce: Human Action Recognition (MS-G3D) Model (Graph Deep Learning)**: Verify & reproduce the findings from the paper. Disentangling and Unifying Graph Convolutions for Skeleton-Based Action Recognition ensuring accuracy & reliability. Uses Disentangled Multi-Scale Aggregation, Disentangling Neighborhoods, and G3D, to effectively model spatial and temporal dependencies within skeletal data.

- **Make It Stand (Computational Fabrication)**: Developed solution for balancing 3D object given a specified point-of-balance, from Make it Stand paper. Draw a plane on the point-of-balance, to carved the object to adjust its CoM directly above the point-of-balance. Discretized the object into voxels (voxelization) withing selected regions that will be empty when printed

- **Conversational ChatBot with Transformers (Deep Learning)**: Developed a Conversational ChatBot utilizing Transformer models in PyTorch, enabling natural and contextually relevant conversations. Leverage the Cornell Movie Dialogues Corpus, containing movie character dialogues.

Personal Projects

Reinforcement Learning for Robotic Arm Manipulation

Tools: PyTorch, MuJoCo, Robosuite

Developed a reinforcement learning agent using the TD3 algorithm to train a robotic arm open a door. Implemented SAC and PPO algorithm and explored its effectiveness for robotic manipulation tasks. Utilized Robosuite on MuJoCo to create realistic physics-based environments for training RL policies.

Multi-Modal Sensor Fusion for Quadrotor Terrain Mapping

Tools: ROS Noetic, Gazebo, PCL, Open3D, RViz, MeshLab

Developed a 3D terrain mapping system for a quadrotor using multi-model sensor (GPS and Depth Sensor),inspired by research in autonomous quadrotor navigation. Implemented sensor fusion algorithm to combine GPS and depth sensor data, for accurate localization and mapping. Utilized ROS Noetic for system integration, Gazebo for simulation, PCL for point cloud processing. Analyzed and redefined 3D terrain models using MeshLab

RL for Safe Drone Autonomy

Tools: ROS Humble, OpenAI Gymnasium, Stable Baselines3, Gazebo

Extended SJTU for autonomous drone navigation system using reinforcement learning in ROS2. Integrated OpenAI gymnasium and stable baselin3 to create a control framework for drone autonomy. Implemented sensor fusion techniques utilizing LIDAR, IMU, GPS data for accurate perception and state estimation. Applied hyperparameter optimization with Optuna & utilized Tensorboard for performance monitoring

Natural Language Robotic Control via LLM

Tools: ROS2, OpenAI, Gazebo, ROSGPT

Control quadrotor using natural language through a fusion of Large Language Models (LLMs) and ROS2. Utilized ROSGPT with quadruped to enable precise, text and voice based execution for seamless human-robot interaction. Translates verbal commands into quadrotor maneuvers using ROSGPT architecture

Attention-Based Vision-Language Models for Captioning

Tools: TensorFlow, OpenCV, Keras

Developed an attention based vision-language model for generating captions for image and video from the Show, Attend and Tell paper. Implemented the model using Tensorflow, Keras for feature extraction and LSTM sequence generation. Utilized large scale dataset such as MSVD and MSR-VTT for training and evaluation. Implemented attention mechanism to focus on the salient part of the visual inputs, enhancing accuracy and relevance of the generated captions.

High-Dimensional Path Planning with RRT

Tools: MuJoCo, RRT algorithm, SciPy

Developed and implemented an enhanced Rapidly-exploring Random Tree (RRT) algorithm for path planning in high-dimensional spaces, Based on RRT: Steven M. LaValle. Optimized standard RRT algorithm, including techniques like, informed sampling, adaptive step sizes, tree rewiring, to improve path quality and planning efficiency.

Scalable Locomotion Policy Learning for Quadruped

Tools: Brax, MuJoCo, Jupyter Notebook

Using policy gradient, particularly PPO and SAC, Developed scalable walking policy for a quadruped, enabling navigation across diverse terrains, inspired from Learning Quadrupedal Locomotion over Challenging Terrain. Integrated Brax and MuJoCo to create realistic environment for training quadruped locomotion policy. Conducted extensive training experiments on Google Colab for efficiently train and evaluate RL models

C3D Optimization for Spatio-Temporal Action Recognition

Tools: TensorFlow, OpenCV, Tensorboard

Build a deep learning system for video action recognition using C3D model, which uses 3D convolutions to understand both space and time in videos, based on the C3D paper. Managed a full space of video analysis, from praparing data to training, testing, deploying the model. Used Tensorflow, OpenCV, Tensorboard to create and fine-tune the C3D model and track model training

Efficient NeRF for Neural Scene Rendering

Tools: TensorFlow, tqdm, Matplotlib, NumPy

Developed an efficient Neural Radiance Fields (NeRF) model, using Tensorflow and Keras, for rendering 3D scenes from 2D images to achieve high-fidelity view synthesis. Applied advanced computer vision and graphics techniques to solve the complex problem of 3D scene reconstruction from sparse 2D views. Focused on developing an efficient varient of NeRF.